

P3 Indp. trials, each with prob. p of success.

P_n = Prob. that n trials result in even number of successes.

$$\text{Aim: } P_n = \frac{1}{2} [1 + (1-2p)^n]$$

Let $A_n = \{n \text{ trials have even number of successes}\}$

$B_n = \{n\text{th trial is a success}\}$

$$\therefore P_n = P(A_n)$$

$$= P(A_n \cap B_n) + P(A_n \cap B_n^c)$$

$$= P(A_{n-1}^c \cap B_n) + P(A_{n-1} \cap B_n^c)$$

$$= P(A_{n-1}^c)P(B_n) + P(A_{n-1})P(B_n^c)$$

$$= (1 - P_{n-1})p + P_{n-1}(1-p)$$

Note:

$$A_n \cap B_n = A_{n-1}^c \cap B_n$$

$$A_n \cap B_n^c = A_{n-1} \cap B_n^c$$

A_{n-1}^c is for 1st, ...,
 $(n-1)$ th trials

B_n is for n th trial

$\therefore$ They are indp.

$$\therefore P_n = (1-2p)P_{n-1} + p$$

① Guess: $P_n = h(1-2p)^n + k \Rightarrow$ Find h and k

② $\rightarrow$ (KKP idea)

$$P_n - p = (1-2p)P_{n-1}$$

$$P_n - p + k = (1-2p)P_{n-1} + k \quad [\text{Try adding constant}]$$

$$\underline{P_n - p + k} = (1-2p) \left(\underline{P_{n-1} + \frac{k}{1-2p}} \right)$$

Choose k so that $\underline{-p+k} = \underline{\frac{k}{1-2p}} =: h \quad \rightarrow \quad \text{We can obtain } k = \frac{2p-1}{2}$

$$\therefore P_n + h = (1-2p)(P_{n-1} + h)$$

Here $Q_n := P_n + h$ is a geometric progression

with $Q_1 = 1-p+h$ and ratio $= 1-2p$

$\Rightarrow$ Obtain $P_n = \dots$